

Coding & Solution

Date	30 September 2026
Team ID	SWTID-2026-3580
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission demonstrates working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure the code is well-structured, commented, and follows best practices for the chosen technology stack.
2. The solution addresses the core problem statement and implements the key functional requirements.
3. All source files, configuration files, and setup instructions needed to run the application are included.
4. The code repository is maintained in GitHub for version control and submission.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/aladiyan33/EduGenie
Programming Language(s)	Python, JavaScript, HTML, CSS
Framework(s) Used	FastAPI, Uvicorn, Jinja2
Key Features Implemented	Q&A, concept explanation, quiz generation, summarization, learning recommendations
Pending / Incomplete Features	Optional future enhancements: voice interaction, multilingual support, gamification, adaptive learning
Setup / Run Instructions	Create virtual environment; install requirements.txt; add GEMINI_API_KEY to .env; run: uvicorn main:app --reload

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

EduGenie is implemented as a FastAPI web application with modular Python AI components and a responsive HTML/CSS/JavaScript interface. The repository contains source code, configuration, tests, templates, static assets, and setup documentation. Gemini is used for Q&A, quizzes, summaries and learning recommendations, while the local explanation module supports concept explanations.
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Implementation Reference:

Repository: github.com/aladiyan33/EduGenie | API routes: /qa, /explain, /quiz, /summarize, /learn/recommendations | Tests: tests/test_api.py and tests/test_modules.py